

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 15E.3

CHROMATOGRAPHY: HPLC AND GC

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	Topic 15: CARBOXYLIC ACID DERIVATIVES, SPECTROSCOPY AND CHROMATOGRAPHY
SUB-TOPIC:	15E.3: Chromatography: HPLC and GC
TOTAL QUESTIONS:	2 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	4 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 15E.3

Q1. 1. (a) Define retention time (t_R) in Gas Chromatography (GC) and High-Performance Liquid Chromatography (HPLC).

[2 Marks]**WCH14/01/Jun23/Q9(a)**

Q2. (b) State how the concentration of a component is determined from a GC chromatogram.

[2 Marks]**WCH14/01/Jun23/Q9(b)**

SECTION B: WORKED MARK SCHEME — 15E.3

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: 1. (a) Define retention time (t_R) in Gas Chromatography (GC) and High-Performance Liquid Chromatography (HPLC). 1. Time taken from injection of sample to detection of maximum peak intensity for a component. 2. Characteristic property under fixed operating conditions.	[2]
Q2	Q2: (b) State how the concentration of a component is determined from a GC chromatogram. 1. Area under the peak (or peak height). 2. Compare with a calibration curve constructed using standard solutions of known concentration.	[2]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 15E.3**FAQ #1: What factors affect retention time in Gas Chromatography?****Category: Chromatographic Parameters****COMMON EXAMINER TRAP / MISTAKE:**

Listing only one factor.

MODEL ANSWER / EXAMINER EXPECTATION:

Retention time depends on (1) boiling point of compound (volatility), (2) solubility/affinity for stationary phase, (3) temperature of column, (4) flow rate of carrier gas.